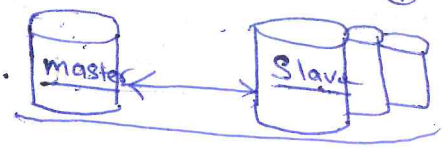
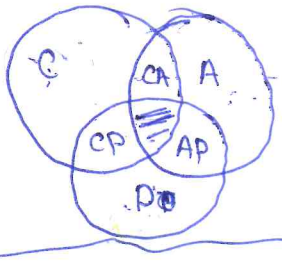
CAP Theorem



* This is a Desirable property of Distributed System with replicated data.

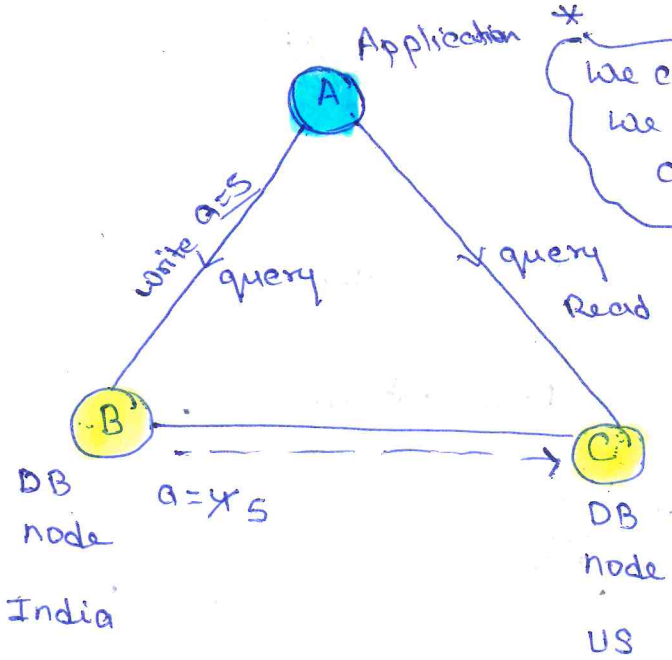
- C - Consistency
- A - Availiability
- P - Partition Talerence

Unique part



*

We can't us ~~three~~ three with. We can use only two CA CP, AP, P.

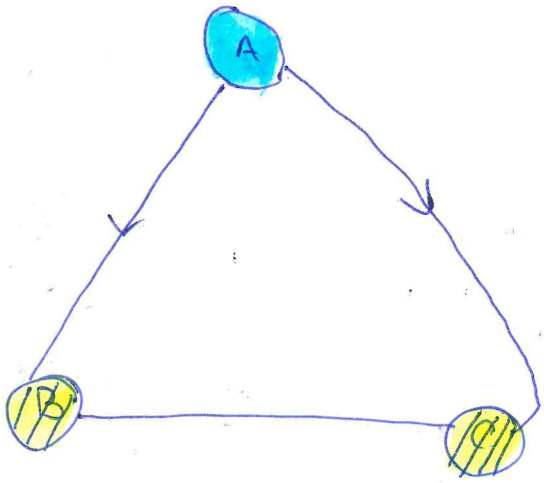


Concept && Coding

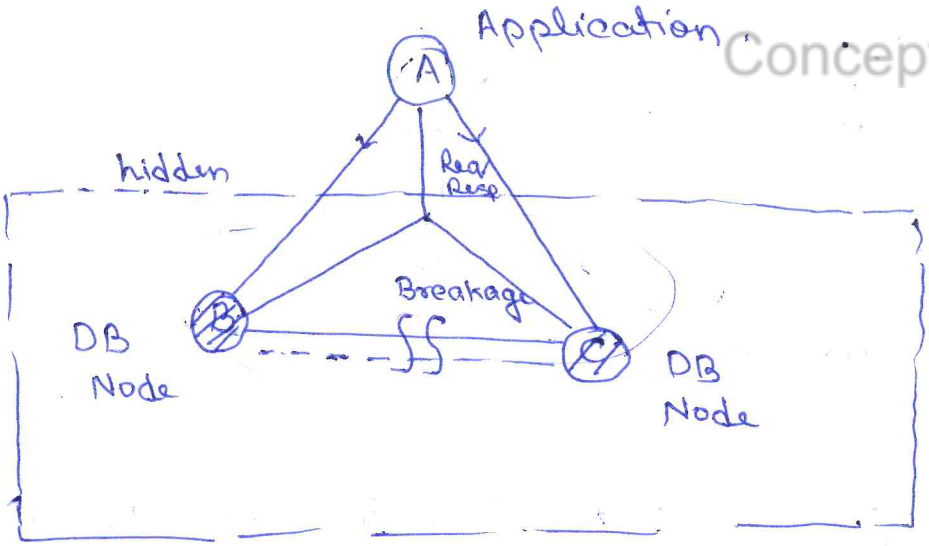
Why ~~C~~ are not use three with.

C - Consistency: → Supose Node B have data a = 4. If Application A write a update query a = 5. B node have update data a = 4 to a = 5. B node have replicate the data on C node. Consitency means A Node is write the data Node B. If A Node Query to Read the C Node. Data always found updated.

A - Availability :- ~~Both Node B an C atleast Response.~~
All Node should response. atleast success or failure.



P - Partition Tolerance :- System is up.



Concept && Coding

* System is hidden. User don't know data is replicated or not. User is only know querying on system.

- Partition Tolerance tell us. If B and C has communication Break. even Application continue quering. that is a Partition Tolerance.

Why CAP Properties Cannot Co-Exist ?

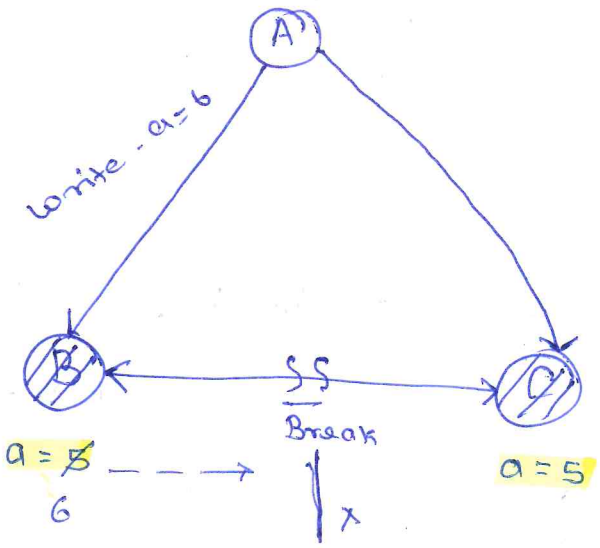
Possible

- ✓ CA – (~~system~~ up)
- ✓ CP –
- ✓ AP –

Not Possible

C A P →
✓ ✓

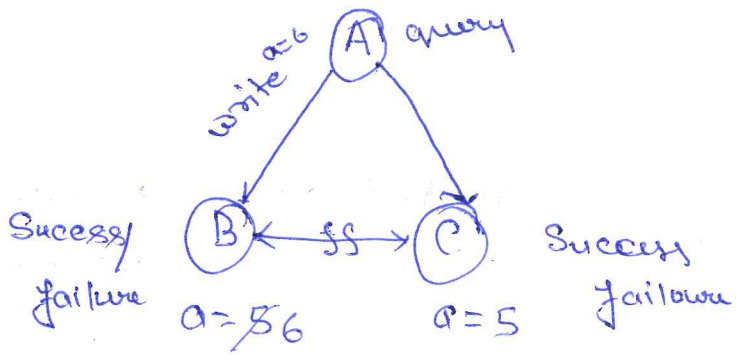
- up
- C
- A



C | A P
x

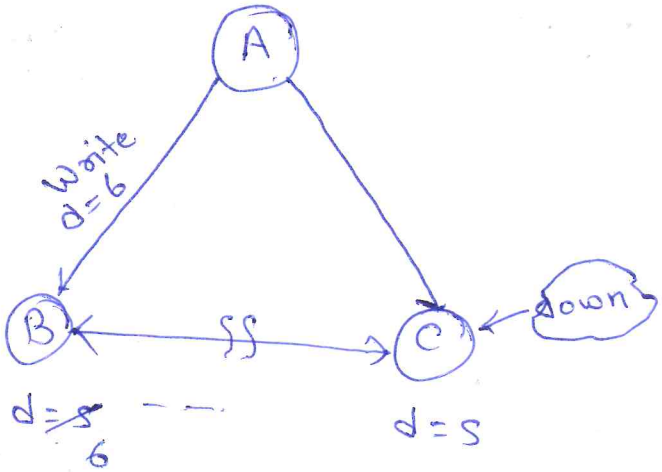
Concept && Coding

Avaliable + Partition Tolerance



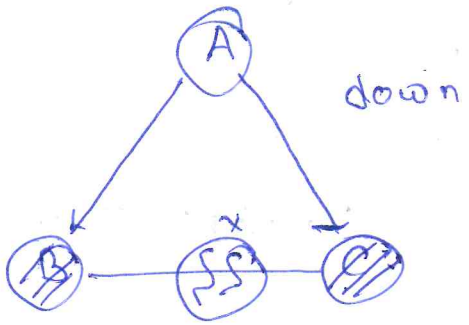
* I do not Consistency loose.

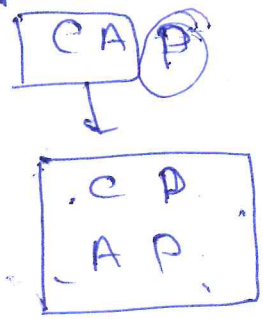
C A P →
~~X~~



* I do Consistency and Availability.

CA P →



Concept && Coding

trade off : we never trade of partition tolerance.